

# Mathematical Mechanics of Range Collapse, Vector Overlap, and Ultra-Episodic Selection in Axomeme 2.0

Technical Diagnostic Analysis

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## Abstract

This document details the mathematical and structural mechanisms behind three core phenomena in deep evolutionary selection transformers: (1) **Epoch 1 Range Collapse** under natural 90:10 prior training and un-converged rate penalty feedback loops, (2) **Vector Projection Overlap** ( $\mathbf{w}^\top \mathbf{h}_j > 0$ ) where high synonymous rates ( $dS \gg 0$ ) trigger false positive LRT calls, and (3) **Ultra-Episodic Single-Branch Selection** ( $p^+ \leq 2\%$ ) where mean species pooling suppresses localized mutational bursts on large trees. We provide formal mathematical proofs, diagnostic failure analyses on *HIV\_RT nex*, and architectural solutions.

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## 1 Root Cause of Epoch 1 Range Collapse

### 1.1 The Feedback Loop of Un-converged Rate Predictions

In early training iterations under a 90:10 natural prior stratification, 90% of all mini-batch training targets are 0.0. The Focal-CORAL Binary Cross-Entropy (BCE) loss applies a strong downward gradient to ordinal threshold logits ( $z_k \rightarrow -\infty$ ).

Before our optimization fix, the biological physical penalty was formulated using the model’s own predictions:

$$\mathcal{L}_{\text{physics\_legacy}} = \text{ReLU}(\hat{y}_{\text{LRT}}) \cdot \text{ReLU}(\hat{\alpha}_{\text{pred}} - \hat{\beta}_{\text{pred}}^+) \quad (1)$$

At Epoch 1, the auxiliary rate heads  $\hat{\alpha}_{\text{pred}}$  and  $\hat{\beta}_{\text{pred}}^+$  are near initialization defaults ( $\hat{\alpha} \approx 1.31, \hat{\beta}^+ \approx 0.50$ ). Consequently,  $\text{ReLU}(\hat{\alpha}_{\text{pred}} - \hat{\beta}_{\text{pred}}^+) > 0$  evaluated to a positive scalar across **100% of training sites** (including true positive selection sites where ground-truth  $dN^+ > dS$ ).

This created a catastrophic feedback loop: un-converged auxiliary rate predictions penalized non-zero LRT predictions ( $\hat{y}_{\text{LRT}} > 0$ ) across every single site in the database. Combined with the 90% neutral prior, all 12 CORAL logits  $z_k$  collapsed to  $-\infty$ , restricting predicted raw LRTs to  $\leq 2.12$ .

## 1.2 The Ground-Truth Rate Target Fix

We resolved this feedback loop by anchoring the physical rate violation penalty to verified ground-truth targets from `meme_results.db`:  $m_{\text{physics}} = \mathbb{I}(y_{\beta_{\text{pos}}, \text{true}} \leq y_{\alpha, \text{true}})$  (2)

$$\mathcal{L}_{\text{physics}} = \frac{1}{B} \sum_{i=1}^B [m_i \cdot (\hat{y}_{\text{LRT}, i})^2 + m_{\text{physics}, i} \cdot (\hat{y}_{\text{LRT}, i})^2] \quad (3)$$

Because  $m_{\text{physics}}$  is a constant derived directly from the database, it never fires on true positive selection sites where  $dN^+ > dS$ , eliminating the feedback loop and preserving dynamic range from Epoch 1 onward.

## 2 Vector Projection Overlap ( $\mathbf{w}^\top \mathbf{h}_j > 0$ )

### 2.1 Embedding Geometry Feature Superposition

Every codon position  $j$  for species  $i$  is embedded as a dual concatenation:

$$\mathbf{x}_{i,j}^0 = \text{Concat}(\mathbf{E}_{\text{codon}}[c_{i,j}], \mathbf{E}_{\text{aa}}[a_{i,j}]) \in \mathbb{R}^{128} \quad (4)$$

Consider three site types:

1. **Invariant Site** (All species TTA / Leucine): All taxa vectors  $\mathbf{x}_{i,j}$  are identical.
2. **High Synonymous Rate Site** ( $dS = 10.0, dN = 0$ , all species Leucine, but using 6 synonymous codons TTA, TTG, CTT, CTC, CTA, CTG): Amino acid tokens are identical, but codon tokens vary significantly across taxa.
3. **Positive Selection Site** ( $dN > dS$ , e.g. L, V, I, M, R, E): Both amino acid tokens and codon tokens vary across taxa.

### 2.2 Species Difference Stream ( $\mathbf{h}_{\text{diff}}$ )

The species pooling layer computes a difference representation:

$$\mathbf{h}_{\text{diff}} = \text{ReLU}(\mathbf{h}_{\text{max}} - \mathbf{h}_{\text{mean}}) \quad (5)$$

On an invariant site,  $\mathbf{h}_{\text{max}} \approx \mathbf{h}_{\text{mean}}$ , so  $\mathbf{h}_{\text{diff}} \approx \mathbf{0}$ . On a positive selection site ( $dN > dS$ ), taxa representations diverge, producing  $\mathbf{h}_{\text{diff}} \gg \mathbf{0}$ .

However, on a high synonymous rate site ( $dS \gg 0$ ), the 6 synonymous codons also cause taxa representations to diverge in codon embedding space ( $\mathbf{E}_{\text{codon}}$ ), producing a large non-zero vector  $\mathbf{h}_{\text{diff}} \gg \mathbf{0}$ .

Because the linear weight vector  $\mathbf{w}$  in `RankConsistentCoralHead` assigns positive weights to variance in  $\mathbf{h}_{\text{diff}}$ , high synonymous rate neutral sites project positively along  $\mathbf{w}$ :

$$\mathbf{w}^\top \mathbf{h}_j > 0 \implies z_0 > 0 \implies \text{False Positive Call} \quad (6)$$

Enforcing  $\mathcal{L}_{\text{physics}}$  penalizes non-zero predictions when  $dN \leq dS$ , forcing  $\mathbf{w}$  to orthogonalize against pure synonymous codon variance.

### 3 Ultra-Episodic Selection ( $p^+ \leq 2\%$ )

#### 3.1 The Species Pooling Dilemma on Large Trees

In HyPhy MEME, episodic selection is modeled via a two-component mixture  $\beta \sim (1 - p^+) \delta_{\beta 1} + p^+ \delta_{\beta +}$ . When  $p^+ \leq 0.02$  on a tree of  $N = 476$  taxa, only 1 to 5 species undergo a non-synonymous substitution, while 471 species carry the identical wild-type amino acid.

When pooling representations across 476 species:

$$\mathbf{h}_{\text{mean}} = \frac{471}{476} \mathbf{u}_{\text{wt}} + \frac{5}{476} \mathbf{u}_{\text{variant}} \approx 0.989 \mathbf{u}_{\text{wt}} \quad (7)$$

The overwhelming background of 471 wild-type species dominates mean and attention pooling ( $\mathbf{h}_{\text{mean}}, \mathbf{h}_{\text{attn}}$ ), projecting a site vector  $\mathbf{h}_j$  that appears 99% identical to a conserved neutral site.

#### 3.2 Architectural Solution: Top- $K$ Extreme Difference Pooling

We resolved this by introducing a Top- $K$  Extreme Difference Pooling stream ( $K = 8$ ):

$$d_i = \|\mathbf{x}_{i,j} - \mathbf{h}_{\text{mean}}\|_2, \quad \mathbf{h}_{\text{top\_k}} = \frac{1}{K} \sum_{k \in \text{TopK}} \mathbf{x}_{k,j} \quad (8)$$

$$\mathbf{h}_j = \text{Linear}(\text{Concat}(\mathbf{h}_{\text{mean}}, \mathbf{h}_{\text{max}}, \mathbf{h}_{\text{attn}}, \mathbf{h}_{\text{diff}}, \mathbf{h}_{\text{top\_k}})) \quad (9)$$

By extracting the mean of the top  $K = 8$  most deviant species representations, the pooling head isolates single-branch mutational spikes regardless of overall tree size  $N$ .